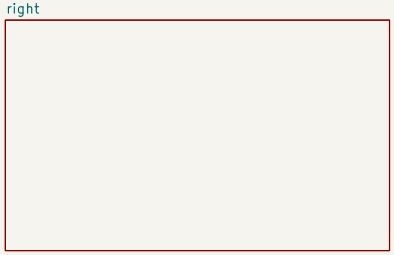
Keygen: one PCB carrying both halves joined by snap-off tabs.
Left half = references 1xx, right half = 2xx. Each half has its own nets (separate GND, VCC).

left



File: left.kicad_sch

right



File: right.kicad_sch

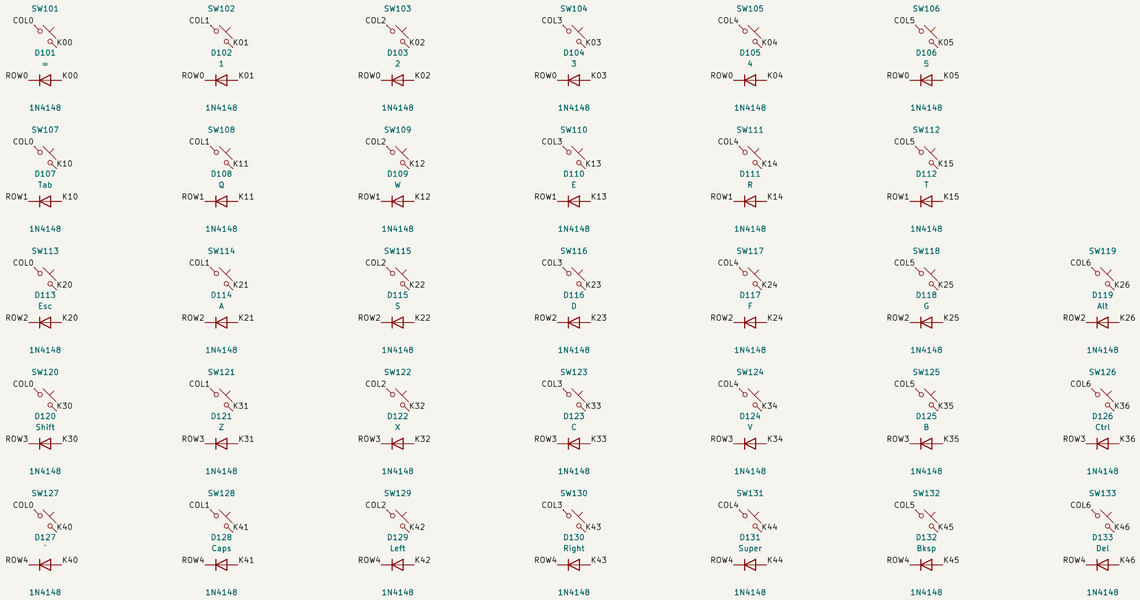
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**github.com/milindmadhukar/keygen**

Sheet: /
File: keygen.kicad_sch

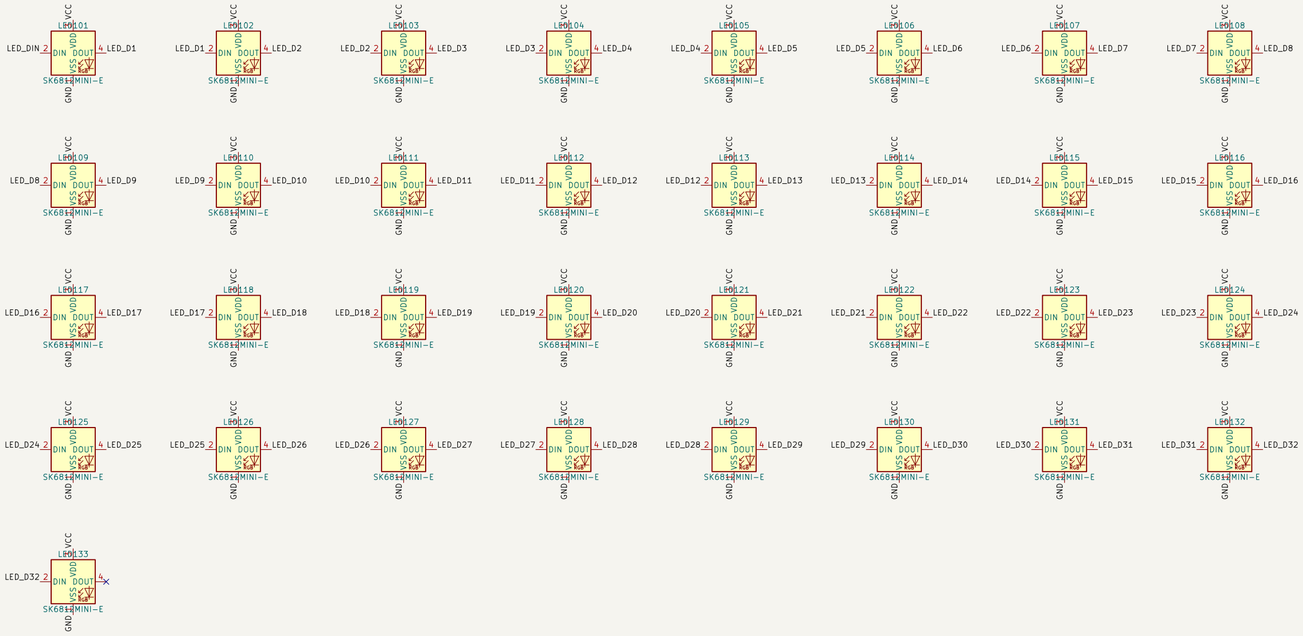
***Title: Keygen***

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|---|---|---|
| Size: A4 | Date: 2026-10-20 | **Rev: 0.1** |
| KiCad E.D.A. 10.0.7 | | Id: 1/3 |

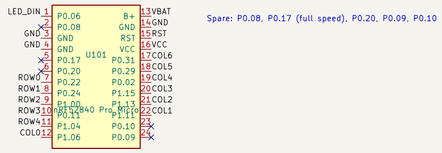
## KEY MATRIX  (COL2ROW: switch to COLn, diode cathode to ROWn)




## PER-KEY LEDs  (33, no underglow) — powered from the controller's switched VCC (P0.13); cap brightness in firmware

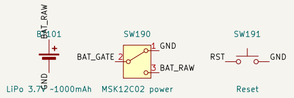



## CONTROLLER  (mounted on the PCB underside, USB-C to the top edge)




Spare: P0.08, P0.17 (full speed), P0.20, P0.09, P0.10

## POWER







C1/C2 sit next to the first LED of the chain.

## BATTERY SWITCH (P-MOSFET; the slide switch only drives the gate)




Switch 2–1: gate to GND, on. Switch 2–3: gate to source, off.
R1 holds the gate off between positions. Leave the switch on while charging:
off, USB charges the cell through the body diode (~0.35 W at 500 mA).

## MOUNTING HOLES (M2)




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Sheet: /left/
File: left.kicad_sch

***Title: Keygen — left half***

| | | |
|---|---|---|
| Size: A2 | Date: 2026-10-19 | **Rev: 0.1** |
| KiCad E.D.A. 10.0.7 | | Id: 2/3 |

## KEY MATRIX  (COL2ROW: switch to COLn, diode cathode to ROWn)

SW201 COL0 K00 D201 – ROW0 K00 1N4148
SW202 COL1 K01 D202 0 ROW0 K01 1N4148
SW203 COL2 K02 D203 9 ROW0 K02 1N4148
SW204 COL3 K03 D204 8 ROW0 K03 1N4148
SW205 COL4 K04 D205 7 ROW0 K04 1N4148
SW206 COL5 K05 D206 6 ROW0 K05 1N4148
Encoder S1/S2 (see ENC201)
D207 ROW0 K06 1N4148

SW208 COL0 K10 D208 \ ROW1 K10 1N4148
SW209 COL1 K11 D209 P ROW1 K11 1N4148
SW210 COL2 K12 D210 O ROW1 K12 1N4148
SW211 COL3 K13 D211 I ROW1 K13 1N4148
SW212 COL4 K14 D212 U ROW1 K14 1N4148
SW213 COL5 K15 D213 Y ROW1 K15 1N4148

SW214 COL0 K20 D214 ' ROW2 K20 1N4148
SW215 COL1 K21 D215 ; ROW2 K21 1N4148
SW216 COL2 K22 D216 L ROW2 K22 1N4148
SW217 COL3 K23 D217 K ROW2 K23 1N4148
SW218 COL4 K24 D218 J ROW2 K24 1N4148
SW219 COL5 K25 D219 H ROW2 K25 1N4148
SW220 COL6 K26 D220 Alt ROW2 K26 1N4148

SW221 COL0 K30 D221 Shift ROW3 K30 1N4148
SW222 COL1 K31 D222 / ROW3 K31 1N4148
SW223 COL2 K32 D223 . ROW3 K32 1N4148
SW224 COL3 K33 D224 , ROW3 K33 1N4148
SW225 COL4 K34 D225 M ROW3 K34 1N4148
SW226 COL5 K35 D226 N ROW3 K35 1N4148
SW227 COL6 K36 D227 Ctrl ROW3 K36 1N4148

SW228 COL0 K40 D228 ] ROW4 K40 1N4148
SW229 COL1 K41 D229 [ ROW4 K41 1N4148
SW230 COL2 K42 D230 Down ROW4 K42 1N4148
SW231 COL3 K43 D231 Up ROW4 K43 1N4148
SW232 COL4 K44 D232 Fn ROW4 K44 1N4148
SW233 COL5 K45 D233 Space ROW4 K45 1N4148
SW234 COL6 K46 D234 Enter ROW4 K46 1N4148

## PER-KEY LEDs  (33, no underglow) — powered from the controller's switched VCC (P0.13); cap brightness in firmware

LED201 LED_DIN LED_D1
LED202 LED_D1 LED_D2
LED203 LED_D2 LED_D3
LED204 LED_D3 LED_D4
LED205 LED_D4 LED_D5
LED206 LED_D5 LED_D6
LED207 LED_D6 LED_D7
LED208 LED_D7 LED_D8
LED209 LED_D8 LED_D9
LED210 LED_D9 LED_D10
LED211 LED_D10 LED_D11
LED212 LED_D11 LED_D12
LED213 LED_D12 LED_D13
LED214 LED_D13 LED_D14
LED215 LED_D14 LED_D15
LED216 LED_D15 LED_D16
LED217 LED_D16 LED_D17
LED218 LED_D17 LED_D18
LED219 LED_D18 LED_D19
LED220 LED_D19 LED_D20
LED221 LED_D20 LED_D21
LED222 LED_D21 LED_D22
LED223 LED_D22 LED_D23
LED224 LED_D23 LED_D24
LED225 LED_D24 LED_D25
LED226 LED_D25 LED_D26
LED227 LED_D26 LED_D27
LED228 LED_D27 LED_D28
LED229 LED_D28 LED_D29
LED230 LED_D29 LED_D30
LED231 LED_D30 LED_D31
LED232 LED_D31 LED_D32
LED233 LED_D32

## CONTROLLER  (mounted on the PCB underside, USB-C to the top edge)

U201 nRF52840 Pro Micro
LED_DIN 1 P0.06
2
GND 3 P0.08
GND 4 GND
5 P0.17
ENC_A 6 P0.20
ROW0 7 P0.22
ROW1 8 P0.24
ROW2 9 P1.00
ROW3 10 P0.11
ROW4 11 P1.04
COL0 12 P1.06
B+ 13 VBAT
GND 14 GND
RST 15 RST
VCC 16 VCC
P0.31 17 COL6
P0.29 18 COL5
P0.02 19 COL4
P1.15 20 COL3
P1.13 21 COL2
P1.11 22 COL1
P0.10 23
P0.09 24 ENC_B

Encoder B on P0.09 (NFC pin): set nfct-pins-as-gpios.
Spare: P0.08, P0.17 (full speed), P0.10

## POWER

BAT201 BAT_RAW GND LiPo 3.7V 1000mAh
SW290 BAT_GATE 2 1 GND 3 BAT_RAW MSK12C02 power
SW291 RST GND Reset

VCC201 10uF GND
VCC202 100nF GND
#FLG201 GND PWR_FLAG
#FLG202 VBAT PWR_FLAG

C1/C2 sit next to the first LED of the chain.

## BATTERY SWITCH (P-MOSFET; the slide switch only drives the gate)

Q201 AO3401A BAT_GATE 1 BAT_RAW 2 VBAT 3
R201 BAT_GATE BAT_RAW 1M

Switch 2-1: gate to GND, on. Switch 2-3: gate to source, off.
R1 holds the gate off between positions. Leave the switch on while charging:
off, USB charges the cell through the body diode (~0.35 W at 500 mA).

## ROTARY ENCODER (EC11) — push switch in the matrix at ROW0/COL6

ENC201 EC11
ENC_A A
GND C
ENC_B B
S1 COL6
S2 K06

## MOUNTING HOLES (M2)

H201 M2
H202 M2
H203 M2
H204 M2
H205 M2

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Sheet: /right/
File: right.kicad_sch

**Title: Keygen — right half**

| | | |
|---|---|---|
| Size: A2 | Date: 2026-10-19 | **Rev: 0.1** |
| KiCad E.D.A. 10.0.7 | | Id: 3/3 |